Recap: **Exercise 6.11**

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{where} \quad \alpha > 0, \quad x \in \mathbb{R}, \quad t > 0,$$

subject to the initial condition $u(x,0) = f(x)$ and $u(x,t) \to 0$ for $x \to \pm\infty$.

We found

$$\widehat{u}(k,t) = \widehat{f}(k)\, \mathrm{e}^{-\alpha\, k^2\, t}$$

and hence

$$u(x,t) = f(x) * \frac{\mathrm{e}^{-x^2\,/\,(4\,\alpha\,t)}}{\sqrt{4\,\pi\,\alpha\,t}}$$

Exercise 6.12: Solve using an LT and an FT (not assessed)

Final Exam

- 3 h in duration
- Worth 65% of your final mark
- 7 questions, 90 marks in total
- At least one question from each of the six topics
- The exam includes a formula sheet (last three pages)
- Taught by Mathematical Physics (2016-2021)
  and then Applied Mathematics (2022-2025)
- 2026 is the first year I have taught it!
  I used the 2025 material with minimal changes.

2024 Q2: Topic 1 (ODE solution existence and reduction of order)

Question 2 (8 marks)

Consider the IVP

$$(t^2 + At + B)\, y''(t) + 2t\, y'(t) - 2y(t) = g(t),$$
$$y(1) = y'(1) = \pi.$$

(a) Let $A = 8, \quad B = 7, \quad g = \frac{e^t}{(2-t)}$.

Find the largest interval D where the IVP is guaranteed a unique solution that is twice differentiable.

Existence and uniqueness of solutions

Theorem

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$$

If f_0, f_1 and g are all continuous over $D = [a, b]$, then over D:

- ODEs: The general solution to $\mathsf{T}y = g$ is a linear sum of 2 linearly independent complementary solutions and a single particular solution.
  - Solutions space (i.e., all complementary solutions) is a 2-dimensional space.
- IVPs: When supplemented with initial values $y(t_0)$ and $y'(t_0)$ where $t_0 \in D$, it has a unique solution which is twice differentiable.

(b) Let $A = B = 0, \quad g = 0.$

Given $v(t) = t$ is a solution, use reduction of order to solve the IVP.

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Given $v(t) = t$ is a solution, use reduction of order to solve the IVP.

Summary of reduction of order

1. Given known complementary solution, $v(t)$, to a second order ODE, write general solution as $y(t) = u(t)v(t)$
2. This reduces the order of the ODE to one: $w' + \left(\frac{2v'}{v} + f_1\right) w = \frac{g}{v}$ where $w = u'$
3. Use the integrating factor $I = v^2 \mathrm{e}^{\int f_1 \mathrm{d}t}$ to solve for w.
4. This leads to the general solution $y = v \int \left[\frac{1}{I} \int \frac{Ig}{v} \mathrm{d}t\right] \mathrm{d}t$

Finding the first solution for $\mathsf{T}v = 0$ is usually the difficult task.

2022 Q3: Topic 1 (ODE systems and diagonalisation)

Question 3 (15 marks)

Consider the following system of ODEs

$$\begin{aligned} x'(t) + 2x(t) - y(t) - 3 &= 0, & x(0) &= 0, \\ y'(t) + 2y(t) - x(t) + 1 &= 0, & y(0) &= 0. \end{aligned}$$

(a) Convert the system into a vector-valued IVP.

(b) Solve the system by matrix diagonalisation.

2022 Q2: Topic 2 (power series and convergence)

Question 2 (15 marks)

(a) Use the Weierstrass test to determine if the following series converges uniformly

$$\sum_{n=0}^{\infty} \frac{2^{-n}\cos(1+n\pi x)}{e^n}.$$

Definition (Uniform convergence of a series)

- Let $W_N(t) = \sum_{n=1}^{N} w_n(t)$ converge uniformly to $W(t)$ on D
- Then $\sum_{n=1}^{\infty} w_n(t)$ converges uniformly to $W(t)$ on D, i.e.,

$$\lim_{N\to\infty} \max_{t\in D} \left| \sum_{n=1}^{N} w_n(t) - W(t) \right| = 0$$

Theorem (Weierstrass test for uniform convergence)

- Let $w_n(t)$ $(n = 1, 2, 3, \ldots)$ be functions on D
- Let $\max_{t\in D} |w_n(t)| \leqslant a_n$ for some $a_n \in \mathbb{R}$ such that $\sum_{n=1}^{\infty} a_n$ converges
- Then $\sum_{n=1}^{\infty} w_n(t)$ converges *uniformly* on D

(b) Consider

$$(1 - x^2)y''(x) - xy'(x) + y(x) = 0.$$

(i) Solve the ODE using a power series expanded about $x = 0$

$$\sum_{n=0}^{\infty} c_n x^n.$$

(ii) State the interval of convergence of any infinite series solutions you find.